O-RAN.WG10.TS.O1Alarms-R005-v01.00

Technical Specification

O-RAN Work Group 10 (OAM for O-RAN)

O-RAN O1 Alarms Specification

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Contents

Foreword

This Technical Specification (TS) has been produced by WG10 of the O-RAN Alliance.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the O-RAN Alarms that are reported over the O1 Interface.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] 3GPP TS 28.111: “Management and orchestration; Fault Management (FM)”

[2] O-RAN.WG10.TS.O1-Interface.0: “O-RAN O1 Interface Specification” (“O1 Interface Specification”)

[3] O-RAN.WG10.TS.Information Model and Data Models.0: “O-RAN Information Model and Data Models Specification” (“IM/DM Specification”)

[4] O-RAN.WG4.TS.MP.0-R005-v20.00: “Management Plane Specification” (“OFH M-Plane Specification”)

[5] O-RAN.WG10.TS.OAM-Architecture: “O-RAN Operations and Maintenance Architecture” (“OAM Architecture Specification”)

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] 3GPP TR 21.905: “Vocabulary for 3GPP Specifications”

3 Definition of terms, symbols and abbreviations

3.1 Terms

Alarm: See Alarm definition in 3GPP TS 28.111 [1].

Error: See Error definition in 3GPP TS 28.111 [1].

Event: See Event definition in 3GPP TS 28.111 [1].

Failure: See Failure definition in 3GPP TS 28.111 [1].

Fault: See Fault definition in 3GPP TS 28.111 [1].

O1 Alarm: An O1 Alarm is an alarm associated with an O-RAN Network Function, that is reported via O1 interface.

Root cause: See Root cause definition in 3GPP TS 28.111 [1].

3.2 Symbols

For the purposes of the present document, the symbols given in 3GPP TR 21.905 [i.1] and the following apply. A symbol defined in the present document takes precedence over the definition of the same symbol, if any, in 3GPP TR 21.905 [i.1].

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [i.1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [i.1].

3GPP	3rd Generation Partnership Project
FM	Fault Management
TR	Technical Report
TS	Technical Specification

4 Overview and General Requirements

4.1 Overview

An O-RAN Network Function may experience faults, which may cause errors that may (or may not) lead to failures, where the function may not deliver its intended service, as specified in 3GPP TS 28.111 [1], clause 4. An O1 Alarm is an alarm associated with an O-RAN Network Function, that is reported via O1 interface.

Fault supervision management services are specified in O1 Interface Specification [2] clause 6.2, and they are responsible for reporting O1 Alarms.

As stated in O1 Interface Specification [2] clause 6.2.0, Stage 2 AlarmList IOC and AlarmRecord data type are specified in 3GPP TS 28.111 [1] clauses 7.3.2 and 7.3.1 and exposed to a Fault Supervision MnS Consumer.

The Alarm Dictionary as specified in IM/DM Specification [3] clause 5.2.1.4.10 and the Alarm Definition as specified in IM/DM Specification [3] clause 5.2.1.4.12 provide additional alarm details beyond what is in the alarm notification, such that the extended data may be used to assist in alarm event processing.

4.2 General Requirements

The general requirements in Table 4.2-1 apply to O1 Alarms.

Table 4.2- 1 General Requirements for O1 Alarms

Requirement label	Description	Motivation
REQ-O1Alarms-MC-1	If an O-DU manages an O-RU in a hierarchical deployment, and if O-DU is able to expose an O-RU Alarm as an Alarm record through the O1 interface, it shall be a compliant O1 Alarm record based on 3GPP TS 28.111 [1].	Identify the requirement for the exposure of an O-RU Alarm through O1 interface of an O-DU as an alarm record in a hierarchical deployment.
REQ-O1Alarms-MC-2	If an O-DU manages an O-RU in a hierarchical deployment, and if O-DU is able to send an alarm from the O-RU as an alarm notification through the O1 interface, that alarm notification shall be a compliant O1 Alarm notification based on 3GPP TS 28.111 [1].	Identify the requirement for the notification of an O-RU Alarm through O1 interface of an O-DU as an alarm notification in a hierarchical deployment.

5 Common Solution Descriptions for O1 Alarms

5.1 O1 Alarm records

An O1 Alarm is described by a set of attributes which is referred to as an O1 Alarm record following the alarm record definition specified in 3GPP TS 28.111 [1], clause 6.3.

The class definition and attributes specified for the alarm record in 3GPP TS 28.111 [1], clauses 7.3.1.1 and 7.3.1.2, respectively, apply to the O1 Alarm record.

5.2 O1 Alarm lists

An O1 Alarm list contains the O1 Alarm records and follows the definition of alarm list specified in 3GPP TS 28.111 [1], clause 6.5. The class definition and attributes specified for the alarm list in 3GPP TS 28.111 [1], clauses 7.3.2.1 and 7.3.2.2, respectively, apply to the O1 Alarm list.

5.3 O1 Alarm Notifications

When an O1 Alarm is raised, or cleared, or modified, the consumers are notified with an O1 Alarm notification, in accordance with the description of alarm notification specified in 3GPP TS 28.111 [1], clause 6.12.

5.4 O1 Alarm Identification

An O1 Alarm is identified using the O1 Alarm Identification. As specified for the alarm identification in 3GPP TS 28.111 [1], clause 6.4, the O1 Alarm records that share the same values for the attributes object instance, alarm type, probable cause and specific problem (if present) are considered to represent the same O1 Alarm in an O1 Alarm List. These four attributes are collectively referred to as the O1 Alarm identifying attributes. The attributes object instance, alarm type, probable cause and specific problem (if present) are also used for O1 Alarm identification in an O1 Alarm notification.

5.5 Correlation of O1 Alarms within the same O1 Fault Supervision MnS producer

A single fault may lead to multiple errors and failures, while a single error may cause multiple failures. In order to capture this relationship among faults, errors, failures and O1 Alarms, as specified for alarm correlation in 3GPP TS 28.111 [1], clause 6.10, O1 Alarm records include rootCauseIndicator attribute and correlatedNotifications attribute. This type of correlation only has a meaning for O1 Alarms from the same O1 Fault Supervision MnS producer.

The modifications to these attributes are reported in an O1 Alarm notification, following the description specified for notifyCorrelatedNotificationChanged in 3GPP TS 28.111 [1], clause 8.7.

NOTE: How to correlate O1 Alarms with alarms from different sources, for example O-Cloud, is out of scope of the present document.

5.6 Acknowledging O1 Alarms

An O1 Alarm record includes attributes (ackState, ackUserId, ackSystemId and ackTime) to enable acknowledgment of an O1 Alarm record, as specified for the alarm records in 3GPP TS 28.111 [1], clauses 6.7 and 7.3.1.2. These attributes are supported if the O1 Fault Supervision MnS producer supports the alarm acknowledgement feature, as specified in 3GPP TS 28.111 [1], clauses 7.3.1.2 and 7.3.1.3. Any change in ackState attribute is notified by notifyAckStateChanged notification, following the specifications for alarm notifications in 3GPP TS 28.111 [1], clauses 6.12 and 8.8.

5.7 Clearing O1 Alarms

The MnS consumer can determine whether an O1 Alarm is automatically or manually cleared, through the attribute of clearingType in AlarmDefinition, included in the AlarmDictionary, as specified in O-RAN.WG10.TS.Information Model and Data Models [3], clauses 5.2.1.4.12.2 and 5.2.1.5.

The perceived severity of an O1 Alarm can be set to “CLEARED” automatically by the O1 Fault Supervision MnS producer if the cause for the alarm is no longer present or no longer detected, as specified for clearing alarms in 3GPP TS 28.111 [1], clause 6.8. The perceived severity of an O1 Alarm can also be set to “CLEARED” manually by the O1 Fault Supervision MnS consumer by setting perceivedSeverity and providing its identity through the attributes of clearUserId and clearSystemId, as specified for clearing alarms in 3GPP TS 28.111 [1], clauses 6.8, 7.3.1.2 and 7.4.1.

Clearing an O1 Alarm is notified by notifyClearedAlarm notification, as specified for alarm notifications in 3GPP TS 28.111 [1], clause 6.12.

The description for active alarms, specified in 3GPP TS 28.111 [1], clause 7.3.1.1 also applies to the active O1 Alarms.

An O1 Alarm that is not active is considered as an inactive O1 Alarm and removed from the O1 Alarm list by the O1 Fault Supervision MnS producer.

The removal of an O1 Alarm from an O1 Alarm list is not directly notified to the O1 Fault Supervision MnS consumer. The removal of an O1 Alarm can be indirectly understood through notifyClearedAlarm and notifyAckStateChanged notifications (depending on whether alarm acknowledgement is supported) as described in 3GPP TS 28.111 [1], clause 6.12.

5.8 Reliability of an O1 Alarm list

The reliability of O1 Alarm list is described in O1 Interface Specification [2], clause 6.2.2.1. The notifications sent by the Fault supervision MnS producer when an O1 Alarm list is not reliable are specified in O1 Interface Specification [2], clause 6.2.1.4.

As specified for the reliability of alarm lists in 3GPP TS 28.111 [1] clause 6.11, clause 7.3.2.2 and clause 7.4.1, unreliableAlarmScope attribute of O1 Alarm list identifies the part(s) of the alarm scope that may not be reliable.

6 Solution Descriptions for Specific O1 Alarms

6.1 O1 Alarm records for Common O-RU Alarms

Common O-RU Alarms are specified in OFH M-Plane Specification [4], Annex A. If an O-DU manages an O-RU in a hierarchical deployment, and if an O-DU is able to expose an O-RU Alarm as an alarm record through the O1 interface, it is exposed as an O1 Alarm record. In this specific case, as specified in OAM Architecture Specification [5], clause 4.2.4.3.1, the O-DU receives a NETCONF based alarm notification from the O-RU, converts the NETCONF based alarm notification to an O1 Alarm notification, updates the alarm records and sends the O1 Alarm notification to the SMO.

As specified in OFH M-Plane Specification [4], Annex A, the common O-RU Alarms are notified to an O-DU by an O-RU using a set of mandatory and optional attributes. To enable O-DU to convert a common O-RU Alarm notification to an O1 Alarm notification, the semantic correspondence of O-RU Alarm attributes to O1 Alarm attributes are provided in Table 6.1-1.

Table 6.1-1: Semantic Correspondence of O-RU Alarms to O1 Alarms

O-RU Alarm attribute as specified in OFH M-Plane Specification [4]	Support Qualifier as specified in OFH M-Plane Specification [4]	Corresponding O1 Alarm attribute	Support Qualifier for O1 Alarm attribute	Explanation
fault-id	M	-	-	The fault-id is specified in OFH M-Plane Specification [4] for O-RU as a type of integer value. The fault-id attribute of an O-RU Alarm does not correspond to any attribute of an O1 Alarm record.
-	-	alarmId	M	The alarmId for an O1 Alarm is specified as a type of string, following 3GPP TS 28.111 [1], clause 7.4.1. The alarmId attribute for an O1 Alarm record does not correspond to any attribute of an O-RU Alarm.

fault-source	M	objectInstance	M	The fault-source is specified in OFH M-Plane Specification [4] for O-RU as a type of string indicating the origin of the alarm. The fault-source in OFH M-Plane Specification [4] may directly reference a defined component or may use a pre-defined textual description. The objectInstance for an O1 Alarm is specified as a type of DN which refers the managed object instance that the alarm is associated with, following 3GPP TS 28.111 [1], clause 7.4.1. See NOTE 1.
fault-severity	M	perceivedSeverity	M	The fault-severity is specified in OFH M-Plane Specification [4] for O-RU as a type of enumeration, with values of CRITICAL, MAJOR, MINOR and WARNING. The is-cleared is specified in OFH M-Plane Specification [4] as a type of Boolean, where the value TRUE states that the O-RU alarm is cleared. WARNING value of the fault-severity attribute of an O-RU Alarm indicates a stateless alarm. The perceivedSeverity of an O1 Alarm is specified as a type of enumeration, with allowed values CRITICAL, MAJOR, MINOR, WARNING, INDETERMINATE and CLEARED, following 3GPP TS 28.111 [1], clause 7.4.1. Clearing of an O1 Alarm is notified by setting perceivedSeverity to CLEARED.
is-cleared	M			
event-time	M	alarmRaisedTime	M	The event-time is specified in OFH M-Plane Specification [4] for O-RU as a type of DateTime to indicate when the fault is detected/cleared. The alarmRaisedTime of an O1 Alarm is specified as a type of DateTime that indicates the date and time the O1 Alarm is raised, following 3GPP TS 28.111 [1], clause 7.4.1. See NOTE 2. The alarmClearedTime of an O1 Alarm is also specified as a type of DateTime that indicates the date and time the O1 Alarm is cleared, following 3GPP TS 28.111 [1], clause 7.4.1. See NOTE 3.
		alarmClearedTime	M	
probable-cause	O	probableCause	M	The probable-cause is specified in OFH M-Plane Specification [4] for O-RU as a type of string, based on ITU-T Rec. X.733. However, the probable-cause is an optional attribute and OFH M-Plane Specification [4] does not specify a standard value for the probable-cause of each common O-RU Alarm. The probableCause of an O1 Alarm is specified as a type of string or integer, following 3GPP TS 28.111 [1], clause 7.4.1 and Annex B. The probableCause of an O1 Alarm is a mandatory attribute.
specific-problem	O	specificProblem	O	The specific-problem is specified in OFH M-Plane Specification [4] for O-RU as a type of string, following ITU-T Rec. X.733 and indicates further information on the alarm than the probable-cause. The specificProblem of an O1 Alarm is specified as a type of string or integer and also provides further refinement to the probableCause of an O1 Alarm, following 3GPP TS 28.111 [1], clause 7.4.1.

proposed-repair-actions	O	proposedRepairActions	O	The proposed-repair-actions is specified in OFH M-Plane Specification [4] for O-RU as a type of string. The proposedRepairActions of an O1 Alarm is specified as a type of string following 3GPP TS 28.111 [1], clause 7.4.1. If the proposed-repair-actions attribute exists, the values reported in the proposed-repair-actions attribute of an O-RU Alarm may also be reported using the proposedRepairActions of an O1 Alarm.
alarm-type	O	alarmType	M	The alarm-type is specified in OFH M-Plane Specification [4] for O-RU as a type of enumeration with specific allowed values. It is an optional attribute. The alarmType of an O1 Alarm is also specified as a type of enumeration with specific allowed values, following 3GPP TS 28.111 [1], clause 7.4.1. It is a mandatory attribute. The allowed values for alarm-type in OFH M-Plane Specification [4] and the allowed values for alarmType in 3GPP TS 28.111 [1] are aligned, with minor differences in string composition, specifically, the use of dashes in alarm-type values and underscores in alarmType values.
fault-name	O	-	-	The fault-name is specified in OFH M-Plane Specification [4] as a type of string to provide name of the fault. This attribute is optional. The fault-name attribute of an O-RU Alarm does not correspond to any attribute of an O1 Alarm record.
fault-text	O	additionalText	O	The fault-text is specified in OFH M-Plane Specification [4] for O-RU as a type of string which indicates a textual description. It is an optional attribute. The additionalText of an O1 Alarm is also specified as a type of string, which allows a free form of text description, following 3GPP TS 28.111 [1], clause 7.4.1. It is also an optional attribute. See NOTE 4.
additional-information	O	additionalInformation	O	The additional-information is specified in OFH M-Plane Specification [4] for O-RU as a type of identifier list which allows the inclusion of additional information set in the alarm, including an identifier (a string) representing the type of the information and an information field (a string) which includes additional information about the alarm. It is an optional attribute. The additionalInformation of an O1 Alarm is specified as a type of NameValuePair list which allows the inclusion of a set of vendor specific alarm information in the alarm, following 3GPP TS 28.111[1], clause 7.4.1. It is an optional attribute.
affected-objects	M	-	-	The affected-objects is specified in OFH M-Plane Specification [4] for O-RU to represent the object or source that is suspected to be affected by this fault. The affected-objects attribute of an O-RU Alarm does not correspond to any attribute of an O1 Alarm record.
NOTE 1: How the fault-source can be captured together with ru-instance-id in a DN is out of scope of this specification.				

NOTE 2: The alarmRaisedTime of an O1 Alarm representing a common O-RU Alarm indicates the time when the O1 Alarm is raised by the O-DU.

NOTE 3: The alarmClearedTime of an O1 Alarm representing a common O-RU Alarm indicates the time when the O1 Alarm is cleared.

NOTE 4: The content for the additionalText of an O1 Alarm can be extended with the value for fault-text attribute of common O-RU Alarm content.

6.2 O1 Alarm representation of Common O-RU Alarms

For O1 Alarms representing common O-RU Alarms the values of the elements fault-id, fault-name, affected-objects, probable-cause and event-time, received from O-RU, shall be included as part of additionalInformation attribute as key value pair in O1 Alarm record, where the key denotes the name and the value denotes value of the element respectively.

For O1 Alarms representing common O-RU Alarms, the values of the *probableCause* and *alarmType* attributes are specified in Table 6.2-1, if the common O-RU alarm has a corresponding probable cause defined in 3GPP TS 28.111 [1].

Table 6.2-1: probableCause and alarmType values for O1 Alarms representing common O-RU Alarms, if the common O-RU alarm has a corresponding probable cause defined in 3GPP TS 28.111 [1]

O-RU Event indicated by O1 Alarm	probableCause (String as defined in 3GPP TS 28.111 [1])	probableCause (Integer as defined in 3GPP TS 28.111 [1])	alarmType
Unit temperature is high, as specified in OFH M-Plane Specification [4], Annex A.	High Temperature	123	ENVIRONMENTAL_ALARM
Unit is dangerously overheating, as specified in OFH M-Plane Specification [4], Annex A.	High Temperature	123	ENVIRONMENTAL_ALARM
Temperature too low, as specified in OFH M-Plane Specification [4], Annex A.	Low Temperature	130	ENVIRONMENTAL_ALARM
Ambient temperature violation, as specified in OFH M-Plane Specification [4], Annex A.	Temperature Unacceptable	350	ENVIRONMENTAL_ALARM
Tuning Failure, as specified in OFH M-Plane Specification [4], Annex A.	Frequency redefinition failed	515	EQUIPMENT_ALARM
Configuring failed, as specified in OFH M-Plane Specification [4], Annex A.	Configuration or Customizing Error	307	PROCESSING_ERROR_ALARM
Critical file not found, as specified in OFH M-Plane Specification [4], Annex A.	File system call unsuccessful	538	PROCESSING_ERROR_ALARM
File not found, as specified in OFH M-Plane Specification [4], Annex A.	File system call unsuccessful	538	PROCESSING_ERROR_ALARM
FPGA SW update failed, as specified in OFH M-Plane Specification [4], Annex A.	Software Error	346	PROCESSING_ERROR_ALARM

C/U-plane logical connection faulty, as specified in OFH M-Plane Specification [4], Annex A.	Connection establishment error	566	COMMUNICATIONS_ALARM
Dying gasp, as specified in OFH M-Plane Specification [4], Annex A.	Power Problem	58	EQUIPMENT_ALARM
Clock source failure, as specified in OFH M-Plane Specification [4], Annex A.	Real Time Clock Failure	70	EQUIPMENT_ALARM
NOTE: Not all common O-RU Alarms that are specified in OFH M-Plane Specification [4], Annex A, are listed in Table 6.2-1.			

For common O-RU Alarms that do not have probable cause mapping in Table 6.2-1, this specification defines the following O-RAN specific probable cause which shall be included as probableCause in O1 alarm notifications. The string value of this probable cause is created by concatenating the string "ORUALARM" with fault-id received in alarm-notif Notification from O-RU on M-Plane as specified in OFH M-Plane Specification [4] clause 11.2 and given in Table 6.2-2. The integer value of this probable cause is created by adding 900 to the value of fault-id received in alarm-notif Notification from O-RU on M-Plane as specified in OFH M-Plane Specification [4] clause 11.2 and given in Table 6.2-2.

Table 6.2-2 O-RAN Specific probable cause for common O-RU Alarms that do not have corresponding probable cause defined in 3GPP TS 28.111 [1]

Probable Cause (String)	Probable Cause (Integer)
ORUALARM{fault-id} See NOTE 1.	9{fault-id} See NOTE 1 and NOTE 2.
NOTE 1: The fault-id is the value of element fault-id, received in alarm-notif Notification from O-RU on M-Plane, as specified in OFH M-Plane Specification [4], Annex A. NOTE 2: An example of integer representation for fault-id 6 is 906.	

6.3 O1 Alarm representation of vendor specific O-RU Alarms

This clause addresses scenario where fault-id values of alarm-notif notification received by O-DU from an O-RU falls into vendor specific range as described in OFH M-Plane Specification [4] clause 11.4.

For vendor specific O-RU Alarms, this specification defines the following O-RAN specific probable cause which shall be included as probableCause in O1 alarm notifications. The string value of this probable cause is created by concatenating the string "ORUALARM" with fault-id received in alarm-notif Notification from O-RU on M-Plane as specified in OFH M-Plane Specification [4] clause 11.4. The integer value of this probable cause is created by adding 900000 to the value of fault-id received in alarm-notif Notification from O-RU on M-Plane as specified in OFH M -Plane Specification [4] clause 11.4.

Table 6.3-1 O-RAN Specific probable cause for vendor specific O-RU Alarms

Probable Cause (String)	Probable Cause (Integer)
ORUALARM{fault-id} See NOTE 1 and NOTE 2.	9{fault-id} See NOTE 1 and NOTE 2.
NOTE 1: An example probable cause string value for a vendor specific O-RU alarm with fault-id 1024 is "ORUALARM1024" and corresponding integer value is 901024. NOTE 2: The representation of the O-RU object (impacted O-RU) is provided as the source of the alarm which allows the SMO to identify the alarm as coming from the O-RU. An SMO would distinguish the semantics of alarms from O-DU versus O-RU with potentially same probable cause value via lookup of appropriate vendor's alarm dictionary. The representation of the O-RU object is not within the scope of this specification.	

6.4 O1 Alarm records for Common O-DU Alarms

The probable cause and alarm type values for O1 Alarm records indicating common O-DU Alarms are provided in Table 6.4-1.

Table 6.4-1: O1 Alarm records indicating common O-DU Alarms

O-DU Event indicated by an O1 Alarm	probableCause (String as defined in 3GPP TS 28.111 [1])	probableCause (Integer as defined in 3GPP TS 28.111 [1])	alarmType
O-DU sync-state changed to HOLDOVER as described in OAM Architecture Specification [5], clause 4.2.8.2. See NOTE.	Clock synchronization problem	506	EQUIPMENT_ALARM
O-DU sync-state changed to FREERUN as described in OAM Architecture Specification [5], clause 4.2.8.2. See NOTE.	Loss of synchronization	518	EQUIPMENT_ALARM
NOTE: These use-cases apply only to O-DU acting as a sync receiver.			

Annex (informative): Change History

Date	Revision	Description
2026.03.08	01.00	Incorporated following approved CRs: - DCM-2024.12.17-WG10-CR-0019-New Clause for Overview-v02.docx - DCM-2025.02.04-WG10-CR-0020 Add Terms, Symbols, Abbreviations and Related References-v03.docx - DCM-2025.02.22-WG10-CR-0021-Add Overview and Related References-v02.docx - DCM-2025.03.24-WG10-CR-0022-Add general requirement and a common description for O1 Alarm Record and O1 Alarm List-v07.docx - DCM-2025.04.28-WG10-CR-0024-Add description for O1 Alarm notification and O1 Alarm identification_v03.docx - DCM-2025.05.30-WG10-CR-0025-Add description for correlation of O1 Alarms-v04.docx - DCM-2025.07.07-WG10-CR-0027-Add description for acknowledging and clearing O1 Alarms-v04.docx - DCM-2025.07.15-WG10-CR-0028-Update description of O1 Alarm records for Common O-RU Alarms-v07.docx - DCM-2025.09.01-WG10-CR-0029-Add description for O1 Alarm List states and reliability-v04.docx - DCM-2025.10.10-WG10-CR-0034-Enhance solution for O1 Alarm records for Common O-RU Alarms-v07.docx - DCM-2025.10.21-WG10-CR-0037-Add Scope-v04.docx - DCM-2025.10.27-WG10-CR-0038-Editorial modifications on references-v01.docx - DCM-2025.10.10-WG10-CR-0033-Add description of O1 Alarm records for Common O-DU Alarms-v05.docx - DCM.AO-2026.01.19-WG10-CR-0041-Add O1 Alarm representation for additional common O-RU Alarms-v08.docx - DCM-2026-01-26-WG10-CR-0042-Update Editors Notes-v03.docx - DCM-2026.02.03-WG10-CR-0043-Clarify alarmRaisedTime and alarmClearedTime in O1 Alarm records-v03.docx - SAM.AO-2026.01.20-WG10-CR-0011- Enhance solution for O1 Alarm representation of Common O-RU Alarms V09.docx - DCM-2026.02.04-WG10-CR-0044-Add O1 Alarm representation of vendor specific O-RU Alarms-v03.docx - DCM-2026.02.19-WG10-CR-0045-Rapporteur's editorial changes-v02.docx

